Q2. [Report] Build a Vehicle Breakdown & Damage Report listing Vehicle ID, Vehicle Type,Driver Name, Breakdown Count, Damage Count, and Total Fuel Cost. Apply Conditional Styling: Vehicles with Breakdown Count > 3 in the month highlighted in Red. Add a calculated Fuel Efficiency column = Distance / Fuel Used. Summarize total breakdown cost impact in the footer. Use Trebuchet MS, size 11.

Objective

Create a Vehicle Breakdown & Damage Report that will:

- Show **Vehicle ID, Vehicle Type, Driver Name**
- Calculate **Breakdown Count and Damage Count**
- Show **Total Fuel Cost**
- Apply conditional styling: Breakdown Count > 3 → Highlight in Red
- Add **Fuel Efficiency column (Distance / Fuel Used)**
- Show **Total Breakdown Cost Impact in Footer**
- Use font **Trebuchet MS, size 11**

Step 1: Understanding the Requirement

First, I understood the problem.The goal was to monitor:

- Vehicle breakdowns
- Damaged shipments
- Fuel usage and cost
- Overall cost impact

Step 2: Preparing the Data

I checked the dataset and ensured it contains:

- Vehicle ID
- Vehicle Type
- Driver Name
- Breakdown Flag (Yes/No)
- Damage Flag (Yes/No)

- Fuel Cost
- Distance
- Fuel Used

These fields are necessary for calculations.

Step 3: Creating the Report in IBM Cognos Analytics

- Opened Cognos Analytics
- Clicked **New Report**
- Selected **List Report**
- Added dataset

Step 4: Grouping the Data

- To analyze data on a monthly basis, I extracted the month from the actual delivery date field and name that new field as month.
- Dragged **Vehicle Type,Driver Name** into the report
- Grouped by **Vehicle Type,Month**

This shows data for each vehicle separately

Step 5: Creating Calculations

- **Breakdown Count**
  sum(case when [Breakdown Flag] = 'Yes' then 1 else 0 end)
- **Damage Count**
  sum(case when [Damage Flag] = 'Yes' then 1 else 0 end)
- **Total Fuel Cost**
  Used aggregation:sum([Fuel Cost])
- **Fuel Efficiency**
  [Distance] / [Fuel Used]

Vehicle ID	Month	Vehicle Type	Driver Name	Breakdown Flag	Damage Flag	Fuel Cost	Distance (km)	Fuel Used (liters)	Fuel Efficiency
<Vehicle ID>	<Month>	<Vehicle Type>	<Driver Name>	<Breakdown Flag>	<Damage Flag>	<Fuel Cost>	<Distance (km)>	<Fuel Used (liters)>	<Fuel Efficiency>
		<Vehicle Type>		<Breakdown Count>	<Damage Count>				
		Total Fuel cost				<Total(Fuel Cost)>			
<Vehicle ID>		<Vehicle Type>	<Driver Name>	<Breakdown Flag>	<Damage Flag>	<Fuel Cost>	<Distance (km)>	<Fuel Used (liters)>	<Fuel Efficiency>
		<Vehicle Type>		<Breakdown Count>	<Damage Count>				
		Total Fuel cost				<Total(Fuel Cost)>			
Total Breakdown and damage count in a month				<Breakdown Count>	<Damage Count>				
Total fuel cost in a month						<Total(Fuel Cost)>			
<Vehicle ID>	<Month>	<Vehicle Type>	<Driver Name>	<Breakdown Flag>	<Damage Flag>	<Fuel Cost>	<Distance (km)>	<Fuel Used (liters)>	<Fuel Efficiency>
		<Vehicle Type>		<Breakdown Count>	<Damage Count>				
		Total Fuel cost				<Total(Fuel Cost)>			
<Vehicle ID>		<Vehicle Type>	<Driver Name>	<Breakdown Flag>	<Damage Flag>	<Fuel Cost>	<Distance (km)>	<Fuel Used (liters)>	<Fuel Efficiency>
		<Vehicle Type>		<Breakdown Count>	<Damage Count>				
		Total Fuel cost				<Total(Fuel Cost)>			
Total Breakdown and damage count in a month				<Breakdown Count>	<Damage Count>				
Total fuel cost in a month						<Total(Fuel Cost)>			
Overall				<Breakdown Count>	<Damage Count>				
Overall - Total						<Total(Fuel Cost)>			

Step 6: Applying Conditional Styling

Applied condition on **Breakdown Count**:

- If Breakdown Count > 3 → Red

This helps identify high-risk vehicles

VEH165			Driver_I	No	Yes	17812.27	1775	180.3	9.84470327
		Bike		1	1				
		Total Fuel cost				30016.86			
VEH032		Truck	Driver_F	No	No	16217.69	1105	154.3	7.16137395
VEH388			Driver_G	No	No	8358.26	820	92.7	8.84573894
VEH070			Driver_I	No	No	14625.18	1634	156.8	10.42091837
VEH104			Driver_J	No	Yes	4959.08	467	55.1	8.47549909
		Truck		0	1				
		Total Fuel cost				44160.21			
VEH326		Van	Driver_A	No	No	20103.59	1355	200.6	6.75473579
VEH417			Driver_C	No	No	19623.63	1759	182	9.66483516
VEH113			Driver_D	Yes	No	17211.44	1875	165.4	11.33615478
VEH221			Driver_D	Yes	No	16515.46	1310	164	7.98780488
VEH218			Driver_F	No	No	13209.97	1581	132.9	11.89616253
VEH386			Driver_F	No	No	10679.83	1231	108.2	11.37707948
VEH247			Driver_G	Yes	No	7803.92	542	79	6.86075949
VEH293			Driver_I	No	No	23852.69	1999	255.9	7.81164517
		Van		3	0				
		Total Fuel cost				129000.53			
Total Breakdown and damage count in a month				4	2				

Step 7: Adding Footer Calculation

Added total breakdown cost impact:

```
total(
  case when [Breakdown Flag] = 'Yes'
  then [Fuel Cost]
  else 0
  end
)
```

Placed this in **footer section**

VEH021	Truck	Driver_B	Yes	No	20943.56	1427	204.8	6.96777344
VEH099		Driver_B	No	No	2627.4	264	24.2	10.90909091
VEH292		Driver_B	No	No	8072.76	972	79.2	12.27272727
VEH119		Driver_E	No	Yes	16397.69	1819	168.6	10.78884935
VEH031		Driver_F	No	No	3954.4	332	41.6	7.98076923
VEH392		Driver_F	No	No	11704.16	940	109.9	8.55323021
VEH454		Driver_F	No	No	24548.7	1683	227.9	7.3848179
VEH331		Driver_J	No	No	10812.02	1257	116.2	10.81755594
	Truck		1	1				
	Total Fuel cost				99060.69			
VEH222	Van	Driver_A	No	No	17127.18	1826	160.3	11.39114161
VEH241		Driver_A	No	No	23819.47	1831	226.3	8.09102961
VEH340		Driver_C	No	No	3728.06	334	38.7	8.63049096
VEH384		Driver_H	Yes	No	18445.05	1791	176.2	10.1645857
VEH402		Driver_H	No	No	22064.19	1782	234.1	7.61213157
VEH037		Driver_I	Yes	No	7148.69	780	79.3	9.83606557

Total Breakdown Cost
501,963.54

Step 8: Formatting

- Selected full report
- Set font:
  - **Trebuchet MS**
  - **Size 11**

Step 9: Final Output

- Verified all calculations
- Checked grouping
- Ensured conditional formatting works

- Confirmed footer shows total cost

Conclusion

By creating this report, I was able to analyze vehicle performance, identify high breakdown vehicles, and calculate the cost impact. This helps improve maintenance planning and reduce operational losses.